MILITARY SPECIFICATION SHEET

# METERS, ITEM TOTALIZING, NON-HERMETICALLY SEALED, ELECTRICAL: TYPE I, 4-40V OR 40-130V OR 10-130V; GRADES A AND B, 2½ INCH AND 3½ INCH, 3 HOLE FLANGE

The complete requirements for procuring meters described herein shall consists of this document and the issue in effect fo specification MIL-M-3971.

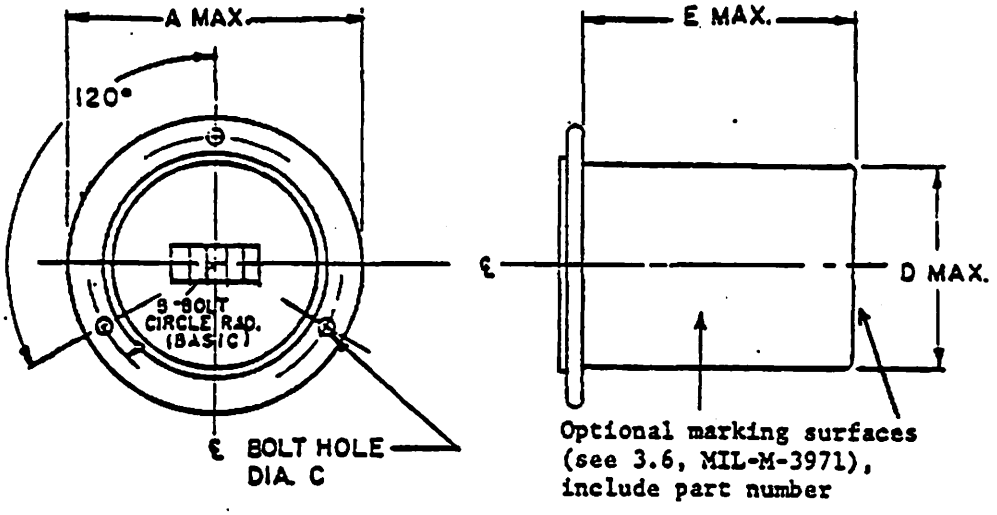



TABLE I

| Dash No. | A | B | C | D | E | Grade | Voltage | Range | Readout |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2.875 | 1.265 | .150 | 2.210 | 3.00 | A | 4-40 | 10,000 | 9999.9 |
| 2 | 2.875 | 1.265 | .150 | 2.210 | 3.00 | A | 40-130 | 10,000 | 9999.9 |
| 3 | 3.500 | 1.580 | .150 | 2.625 | 3.00 | A | 4-40 | 10,000 | 9999.9 |
| 4 | 3.500 | 1.580 | .150 | 2.625 | 3.00 | A | 40-130 | 10,000 | 9999.9 |
| 5 | 2.875 | 1.265 | .150 | 2.210 | 3.00 | B | 4-40 | 10,000 | 9999.9 |
| 6 | 2.875 | 1.265 | .150 | 2.210 | 3.00 | B | 40-130 | 10,000 | 9999.9 |
| 7 | 3.500 | 1.580 | .150 | 2.625 | 3.00 | B | 4-40 | 10,000 | 9999.9 |
| 8 | 3.500 | 1.580 | .150 | 2.625 | 3.00 | B | 40-130 | 10,000 | 9999.9 |
| (A) 9 | 2.810 | 1.580 | .150 | 2.135 | 1.75 | B | 10-130 | 10,000 | 9999.9 |
| (A) 10 | 3.500 | 1.580 | .150 | 2.135 | 1.75 | B | 10-130 | 10,000 | 9999.9 |



FSC 6645

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NOTES:

1. Dimensions are in inches
2. Tolerances: Dimensional ± 0.015 inches, unless otherwise indicated
   Angular ± 0.5°
3. Range and readout are in hours

REQUIREMENTS

Dimensions of case: See Figure and Table I.

Colors, elapsed time indicator:

Hours: White on black background
Tenths: Black on white background

Power consumption: Not applicable

Dielectric strength:

| | | |
|---|---|---|
| Voltage | - | 200 volts rms |
| Frequency | - | 60 Hz |
| Potential | - | ac |
| Points of application | - | Terminals and case |

Weight: 0.60 lbs maximum

Starting:

| | | |
|---|---|---|
| 4-40V: | 1. | Energize with 28V for 60 seconds at +22°C (±2°C), de-erergize for 180 seconds, repeat 3 times. Meter to start within 10 seconds. |
| | 2. | As above at 12V, repeat 2 times. |
| 40-130V: | 1. | As above at 120V, repeat 3 times. |
| | 2. | As above at 85V, repeat 2 times. |
| (A) 10-130V: | 1. | As above at 120V, repeat 3 times. |
| | 2. | As above at 85V, repeat 2 times. |
| | 3. | As above at 28V, repeat 3 times. |
| | 4. | As above at 12V, repeat 2 times. |

Stopping: Within 120 seconds after being de-energized.

Power supply tolerance:

| | | |
|---|---|---|
| 4-40 voltmeters | - | ½ hour at 4 volts |
| | - | ½ hour at 40 volts |
| 40-130 voltmeters | - | ½ hour at 40 volts |
| | - | ½ hour at 130 volts |
| (A) 10-130 voltmeters | - | ½ hour at 10 volts |
| | - | ½ hour at 130 volts |

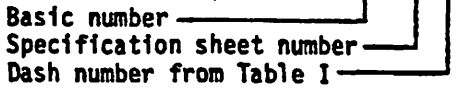
Part number: Consists of the basic number of this specification sheet and a dash number taken from Table I.

(A) denotes changes



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Example: M3971/1-1




Supersession data:

MS 16970-1 replaced by M3971/1-(5 or 6)
MS 16970-2 replaced by M3971/1-(7 or 8)

Custodians:
Army - AR
Navy - SA
Air Force - 99

Review activities:
Army -
Navy - YD
Air Force - 11

User activities:
Army - ER, ME
Navy - MC, AS

Preparing activity:
Army - AR

Project No. 6645-0344



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